

Understanding Unmet Medical Need in Europe: The Role of Individual and Macrosocial Determinants

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Abstract

Background and Objectives: Even though universal healthcare systems attempt to reduce systemic barriers to access to care, unmet medical need remains widely prevalent across and within countries. This study attempts to deepen our understanding of the role of individual-level sociodemographic and macro-level welfare regime variables in determining self-reported unmet need for medical care among Europeans.

Methods: We used survey data from 30 European countries from 2014 Round 7: Social Inequalities in Health of the European Social Survey (ESS7). Using Bayesian Binary and Multilevel Logistic Regressions in R Studio, we explored the effects of sociodemographic determinants of unmet medical need in the past 12 months, as well as the potential cross-country and cross-welfare regime variation in outcomes. We estimated a set of four weighted Bayesian logistic regression models, two of which were fixed effect models with full pooling and two of which were conducted in a multilevel framework and used partial pooling. Model fit was compared using the WAIC functions.

Results: We found that smoking status, gender, age, education, ethnic minority status and self-reported health are consistently predicting the odds of unmet medical need, and the magnitude of their effects does not change across models. Specifically, we found that regular and past smokers, females, younger and more educated individuals, ethnic minorities, and those who report worse health outcomes are more likely to have faced unmet medical need in the past 12 months, compared to non-smokers, males, older and less educated individuals, those who do not self-identify as ethnic minorities and who report good health, respectively. Controlling for welfare regimes initially slightly improved our model fit, but after including country fixed effects, the differences in unmet need initially attributed to welfare regimes disappeared. Instead, our data showed significant variation in unmet need for care across the 30 European countries.

Conclusions: Our findings shed light on important risk factors associated with unmet need for medical care. Since unmet need is not uniformly distributed across the population in Europe, policy makers should consider investing in healthcare accessibility programs within a ‘targeted-universalist’ framework, whereby subpopulations at higher risk of unmet need are targeted with additional support within a universal healthcare policy system.

Keywords: *Unmet medical need, health care access barriers, welfare regimes, Bayesian logistic regressions, European Social Survey*

1 Background and Rationale

Even though universal healthcare systems attempt to reduce systemic barriers to access to care, unmet medical need remains widely prevalent across and within countries. This perceived inability to access healthcare services can arise both as a feature of broader policy systems on a macro-level, or as a function of individual-level characteristics, choices, or constraints (Hwang 2018). To date, however, our understanding of the interaction between systemic and individual determinants of unmet need for care remains limited. Existing studies have predominantly focused on individual-level determinants of subjective unmet need (SUN), (Yoon et al. 2019), or limited their scope of analysis to a national level (Jung and Ha 2021), without focusing on how individual and shared systemic factors may interact and reinforce each other.

Broadening the scope of analysis by using a large multi-national dataset, such as European Social Survey (ESS), will allow us to identify common systemic policy features across countries. At the same time, it will provide us with a large-enough analytical sample to justify modeling macro- and micro-level interactions in a multilevel Bayesian framework. Since Europeans live in their respective countries with distinct, yet often shared policy characteristics and healthcare system outcomes (i.e., number of hospital beds per capita, sickness leave generosity), in a multinational pooled dataset, it is important to be able to control for differences and similarities in policy design in order to produce a more precise set of estimates for determinants of unmet medical need. This study aims to leverage a large multinational dataset, ESS7 (N = 39,891), to shed light on how specific cross-country differences affect unmet medical need.

Research Question

Do individual-level sociodemographic and macro-level policy variables determine self-reported unmet need for medical care among Europeans?

Hypothesis

Based on a preliminary review of literature, we hypothesize that the likelihood of unmet medical need is associated with lower socioeconomic status and that our proxy measures of education, ethnic minority status and employment flexibility will help us determine whether this is the case. We also hypothesize that the macropolitical environment in which individuals live, including the different distinct welfare regime configuration, matters as a determinant of unmet medical need in Europe. Lastly, we expect significant cross-country variation in unmet medical need.

2 Methodology

Data

We use the 2014 Round 7: Social Inequalities in Health of the European Social Survey (ESS7), a bi-annual, cross-sectional international survey of 30 European countries. The ESS7 involves rigorous translation protocols for maximum cross-country comparability and is conducted using hour-long face-to-face CAPI interviews (European Social Survey 2018). The ESS7 dataset is publicly available, encrypted, and anonymized. The ESS7 frame for survey is based on strict random probability sampling, representative of all persons aged 15 and over in Europe with an effective minimum sample size of at least 800 individuals (European Social Survey 2018). The survey uses a balanced response rate strategy across countries, with 70% representing the minimum target response rate in each country surveyed, and to adjust for unequal representation of subgroups, post-stratification and population-level weights are provided and used in this analysis.

Target population and definition of analytic sample

The target population for this analysis are all European individuals aged 15 and older at risk of having faced unmet medical need in the past 12 months. By survey design, ESS7 included only individuals aged 15 and older, and so no age restrictions on the dataset needed to be made. In our analytic sample, we excluded individuals with missing responses for our main outcome variable of interest, a binary ‘yes/no’ question for whether the respondent faced unmet need for medical care in the past 12 months. Only complete cases with respect to all main variables were included in the analysis and missing values were uniformly recoded and excluded using listwise deletion. We have not decided to impute missing data, since overall missingness for our main variables was less than 1%. From the total of 39,891 ESS7 respondents, we excluded 5,862 individuals, or 14.7% of the total sample based on our exclusion criteria, yielding an analytic sample of 34,029 unique respondents.

Analytic method

To estimate the likelihood of having faced unmet medical need in the last 12 months prior to the interview, we used Bayesian logistic regression with Bernoulli distribution to estimate 6 models weighted by survey and population weights. Three of the models were fixed-effect models with full pooling, and the outstanding three were multi-level models with partial pooling. We set non-informative priors for our models. In Model 5 and 6, welfare regimes, while considered as second-level terms, have been implemented as first-level components for computational efficiency. The models are specified as follows:

Model 1: Individual-level determinants of unmet medical need (Fixed effects model, Bernoulli distribution, full pooling) — age, gender, ethnicity, self-reported health, smoking history, unemployment, education years, work flexibility, marital status, happiness and government satisfaction.

Model 2: Model 1 + welfare regime classification (Fixed effects, full pooling).

Model 3: Model 2 + country-fixed effects (Fixed effects, full pooling).

Model 4: Model 1 + welfare regime classification on the second level (Multilevel, partial pooling).

Model 5: Model 4 + country effects on the second level (Multilevel, partial pooling).

Model 6: Model 5 + random slopes by ethnicity (Multilevel, partial pooling).

Apart from providing our results as exponentiated coefficients in table forms with credible intervals, our findings will also be presented graphically using density plots with Markov-Chain Monte Carlo draws from the posterior distributions.

Variables

The main outcome of interest is a binary variable for the risk of facing unmet need for medical care during the last 12 months, based on the question: ‘In the last 12 months, were you unable to get medical consultation or treatment?’.

The independent variables in this analysis consisted of 11 individual-level sociodemographic variables, 1 macro-level welfare regime variable, and 1 variable for country-fixed effects. Our individual-level variables include: a dummy variable for self-reported gender; a categorical variable for self-reported health; a dummy variable for self-reported ethnic minority status in the country of residence; a continuous, standardized variable for the number of years of a respondent’s full-time education; a categorical variable for smoking history; a continuous standardized variable for the age of respondent at the time of interview; a continuous variable for self-reported workplace flexibility in how daily work is organized; a dummy variable for whether an individual is currently unemployed; a continuous variable for the level of respondents’ satisfaction with the government (1 = extremely dissatisfied to 10 = extremely satisfied); a continuous variable for the level of respondents’ self-reported happiness (1 = extremely unhappy

to 10 = extremely happy); a categorical variable for marital status.

These sociodemographic covariates were chosen with reference to established research evidence on individual-level determinants of unmet medical need (Hawks et al. 2020; Hwang 2018; Moran et al. 2021; Wiltshire et al. 2009).

Lastly, we have computed and coded a macro-level welfare explanatory variable to classify respondents' countries of residence based on welfare regime categorization. More specifically, we have created a categorical variable for welfare regime type, based on an adapted regime typology, including elements of Esping-Andersen's original three world typology (Esping-Andersen 1990), Ferrera's additions (Ferrera 1996) and Eikemo's modifications (Eikemo et al. 2008), such that 5 distinct regimes were identified, the Scandinavian, Continental/Conservative, Mediterranean, Central/Eastern European and Anglo-Saxon (see Appendix A for detailed classification of each country included). In addition, since Israel could not be classified within either of the regimes, we have decided to include it as a separate regime category.

Welfare regime typology literature suggests that individual regimes are based on different reliance on the state-family-market distribution of services and responsibilities, whereby some regimes, such as the Conservative Regime, rely more heavily on families for provision of care, than other regimes, such as the Scandinavian, where care is more outsourced to the responsibility of the state (Aspalter 2020). Using welfare regimes is particularly useful because it allows us to categorise countries according to 'ideal types' of welfare institution design, which allows us to take advantage of 6 informative categories of regimes with similar cross-country welfare provision attributes. This offers us valuable potential comparative insights into the effect particular designs of welfare provision have on the uptake of caring responsibilities in families and health inequalities in our societies.

3 Results

Table 1: Descriptive Statistics of Analytic Sample by unmet medical need categories

Characteristic	0, N = 29,777 ¹	1, N = 4,252 ¹
Age (years)	51 (37, 64)	47 (34, 61)
Gender		
0 Female	15,168 (51%)	2,549 (60%)
1 Male	14,609 (49%)	1,703 (40%)
Self-reported health		
1 Very Poor	309 (1.0%)	117 (2.8%)
2 Poor	1,554 (5.2%)	450 (11%)
3 Good	7,334 (25%)	1,456 (34%)
4 Very Good	13,237 (44%)	1,641 (39%)
5 Excellent	7,343 (25%)	588 (14%)
Ethnic minority		
0 No	28,119 (94%)	3,911 (92%)
1 Yes	1,658 (5.6%)	341 (8.0%)
Years of education	13.0 (11.0, 16.0)	13.0 (11.0, 16.0)
Smoker history		
1 Never smoked	12,755 (43%)	1,609 (38%)
2 Past smoker	9,884 (33%)	1,401 (33%)
3 Current smoker	7,138 (24%)	1,242 (29%)
Work organization Flexibility	7.0 (4.0, 9.0)	7.0 (4.0, 9.0)
Unemployment		
0 No	17,259 (58%)	2,450 (58%)
1 Yes	12,518 (42%)	1,802 (42%)
Satisfied with government	5.00 (3.00, 6.00)	4.00 (2.00, 5.00)
Self-reported happiness	8.00 (7.00, 9.00)	7.00 (6.00, 8.00)
Marital status		
1 Married/civil union	16,160 (54%)	2,079 (49%)
2 Never married	7,772 (26%)	1,289 (30%)
3 Divorced/Separated	3,373 (11%)	570 (13%)
4 Widowed	2,472 (8.3%)	314 (7.4%)
Welfare regime		
1 Conservative	9,452 (32%)	1,169 (27%)
2 Scandinavian	5,399 (18%)	781 (18%)
3 Mediteranean	2,022 (6.8%)	377 (8.9%)
4 Central/Eastern EU	7,747 (26%)	1,128 (27%)
5 Liberal	3,551 (12%)	432 (10%)
6 Israel	1,606 (5.4%)	365 (8.6%)
¹ Median (IQR); n (%)		

Table 1 above shows the summary statistics, including the median with interquartile range for continuous variables, as well as the number of observations with the percentage of overall sample for categorical variables, as stratified by our binary outcome variable – unmet need for care. The median age was four years lower for those who faced unmet need compared to those who did not. We see that we have an even gender distribution among those who did not face unmet need for care, but that 60% of those who faced unmet need in the past 12 months were female, compared to 40% of males. Poor self-reported health outcomes are also much more common among those who faced unmet need. Ethnic minorities form less than 9% of the overall analytic sample both among those who did and did not face unmet medical need. Median education duration, thirteen years, was the same among both groups. Current smokers and unemployed individuals make up respectively less than 30% and 42% of the analytical sample in both groups. On average, those who face unmet need for care are less satisfied with their government and are more likely to have never married or divorced.

In the tables (Table 2–7) and figures (Figure 1–8) below, we present the results from our 6 Bayesian logistic regression models, as specified previously.

Table 2: Model 1 - Individual-level determinants of unmet medical need (Bernoulli fixed effect model)

	Estimate	Est.Error	Q2.5	Q97.5
Intercept	0.6916	1.143	0.5301	0.9027
Male (Female)	0.6982	1.036	0.6513	0.7482
Fair health (Poor)	0.8077	1.133	0.6331	1.03
Good health (Poor)	0.5441	1.125	0.4313	0.6815
Very good health (Poor)	0.2985	1.127	0.2366	0.3784
Excellent health (Poor)	0.1781	1.136	0.1391	0.227
Ethnic minority (Poor)	1.352	1.066	1.188	1.528
Education years	1.092	1.018	1.054	1.132
Age	0.6924	1.025	0.6597	0.7265
Past smoker (non-smoker)	1.181	1.041	1.091	1.28
Current smoker (non-smoker)	1.166	1.045	1.07	1.269
Work Flexibility	1.029	1.005	1.018	1.039
Unemployment (no)	1.049	1.04	0.9727	1.133
Government satisfaction	0.9258	1.008	0.9113	0.9405
Happiness	0.9425	1.01	0.9252	0.9602
Never married (Married)	1.007	1.048	0.9218	1.102
Divorced/Separated (Married)	1.113	1.055	1.002	1.236
Widowed (Married)	1.013	1.074	0.8791	1.161

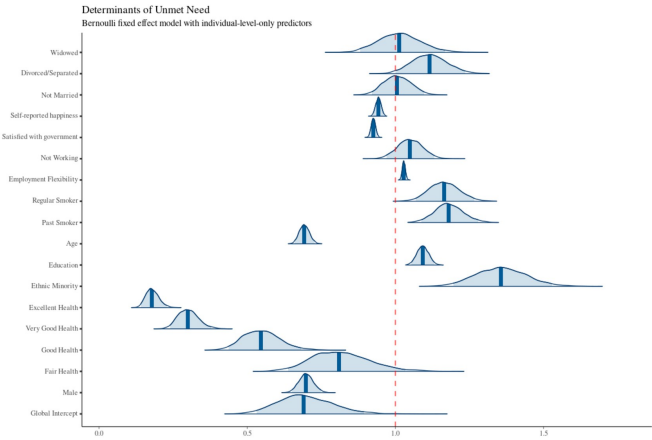


Figure 1: Individual-level determinants of unmet medical need (Bernoulli fixed-effect logistic regression)

Table 3: Model 2 - Individual-level determinants of unmet medical need and welfare regimes (Bernoulli fixed effect model)

	Estimate	Est.Error	Q2.5	Q97.5
Intercept	0.6372	1.148	0.4883	0.8377
Male (Female)	0.6917	1.036	0.6446	0.7418
Fair health (Poor)	0.8516	1.132	0.6705	1.088
Good health (Poor)	0.5795	1.124	0.4603	0.728
Very good health (Poor)	0.3186	1.126	0.2524	0.402
Excellent health (Poor)	0.174	1.135	0.1349	0.2216
Ethnic minority (Poor)	1.299	1.066	1.142	1.467
Education years	1.093	1.018	1.055	1.134
Age	0.6944	1.025	0.6619	0.7293
Past smoker (non-smoker)	1.2	1.043	1.108	1.305
Current smoker (non-smoker)	1.186	1.045	1.089	1.294
Work Flexibility	1.025	1.005	1.014	1.036
Unemployment (no)	1.033	1.039	0.9591	1.11
Government satisfaction	0.9289	1.008	0.9144	0.9432
Happiness	0.9335	1.01	0.9151	0.9519
Never married (Married)	1.008	1.048	0.919	1.103
Divorced/Separated (Married)	1.121	1.057	1.005	1.243
Widowed (Married)	1.038	1.075	0.9013	1.193
Scandinavian (Conservative)	1.301	1.052	1.181	1.439
Mediterranean (Conservative)	1.239	1.069	1.09	1.412
Central/Eastern EU (Conservative)	0.9766	1.048	0.8907	1.07
Liberal (Conservative)	1.05	1.062	0.9297	1.177
Israel (Conservative)	2.068	1.073	1.804	2.378

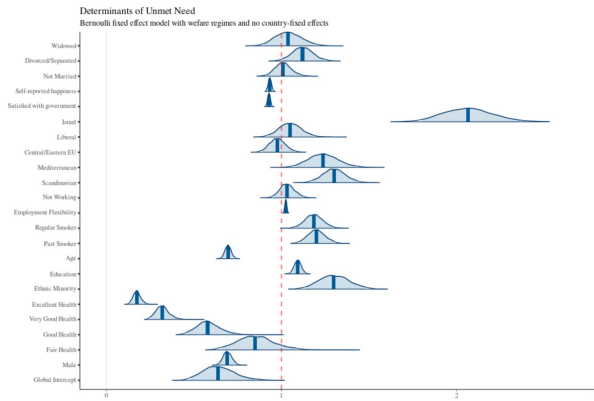


Figure 2: Individual-level determinants of unmet medical need and welfare regimes (Bernoulli fixed-effect logistic regression)

Model 1 includes only individual-level predictors (Table 2, Figure 1). Model 2 adds our welfare regime classification (Table 3, Figure 2): compared to the Conservative regime, respondents living in the Scandinavian and Mediterranean regimes, as well as in Israel, showed higher odds of unmet medical need.

Table 4: Model 3 - Individual-level determinants of unmet medical need, welfare regimes and country-fixed effects (Bernoulli fixed effect model)

	Estimate	Est.Error	Q2.5	Q97.5
Intercept	0.3573	1.185	0.2559	0.4974
Male (Female)	0.6824	1.037	0.6362	0.7316
Fair health (Poor)	0.8473	1.134	0.6621	1.079
Good health (Poor)	0.5752	1.123	0.4601	0.7225
Very good health (Poor)	0.3342	1.126	0.2642	0.4222
Excellent health (Poor)	0.1926	1.134	0.1509	0.2479
Ethnic minority (No)	1.233	1.069	1.082	1.406
Education years	1.085	1.019	1.045	1.126
Age	0.6982	1.025	0.6655	0.7329
Past smoker (non-smoker)	1.179	1.043	1.087	1.28
Current smoker (non-smoker)	1.186	1.045	1.09	1.29
Work Flexibility	1.015	1.005	1.004	1.025
Unemployment (no)	1.031	1.041	0.9538	1.114
Government satisfaction	0.9429	1.009	0.9272	0.9591
Happiness	0.9222	1.01	0.9047	0.9397
Never married (Married)	0.9982	1.048	0.9089	1.096
Divorced/Separated (Married)	1.155	1.057	1.035	1.292
Widowed (Married)	1.031	1.072	0.9009	1.179
Scandinavian (Conservative)	1.904	1.6	0.754	4.841
Mediterranean (Conservative)	1.812	1.789	0.6064	5.786
Central/Eastern EU (Conservative)	1.553	1.46	0.7413	3.31
Liberal (Conservative)	1.503	1.81	0.4604	4.766
Israel (Conservative)	1.995	1.989	0.5075	7.719
Belgium (Austria)	1.794	1.146	1.375	2.347
Switzerland (Austria)	1.372	1.169	0.9968	1.871
Czechia (Austria)	0.723	1.455	0.3446	1.485
Germany (Austria)	2.853	1.125	2.261	3.607
Denmark (Austria)	0.6608	1.598	0.2683	1.676
Estonia (Austria)	1.699	1.453	0.8271	3.448
Spain (Austria)	1.14	1.781	0.3705	3.471
Finland (Austria)	1.961	1.594	0.7806	4.921
France (Austria)	3.103	1.129	2.432	3.937
Great Britain (Austria)	1.74	1.816	0.5403	5.634
Hungary (Austria)	0.4739	1.469	0.2244	1.011
Ireland (Austria)	0.9033	1.819	0.281	3.038
Israel (Austria)	1.978	1.986	0.5175	7.767
Lithuania (Austria)	1.094	1.452	0.5244	2.253
Netherlands (Austria)	0.715	1.165	0.5261	0.9586
Norway (Austria)	1.408	1.598	0.5609	3.568
Poland (Austria)	2.603	1.453	1.258	5.319
Portugal (Austria)	1.671	1.778	0.5366	5.009
Sweden (Austria)	1.042	1.596	0.4205	2.65
Slovenia (Austria)	0.8242	1.46	0.3865	1.704

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Model 3 (Table 4) adds country-fixed effects to the welfare-regime model. Once country effects are included, the welfare-regime coefficients are attenuated and lose precision, while substantial country-to-country heterogeneity emerges (for example, Germany, France, and Poland show markedly higher odds of unmet need relative to the Austria reference category, while Hungary and the Netherlands trend lower).

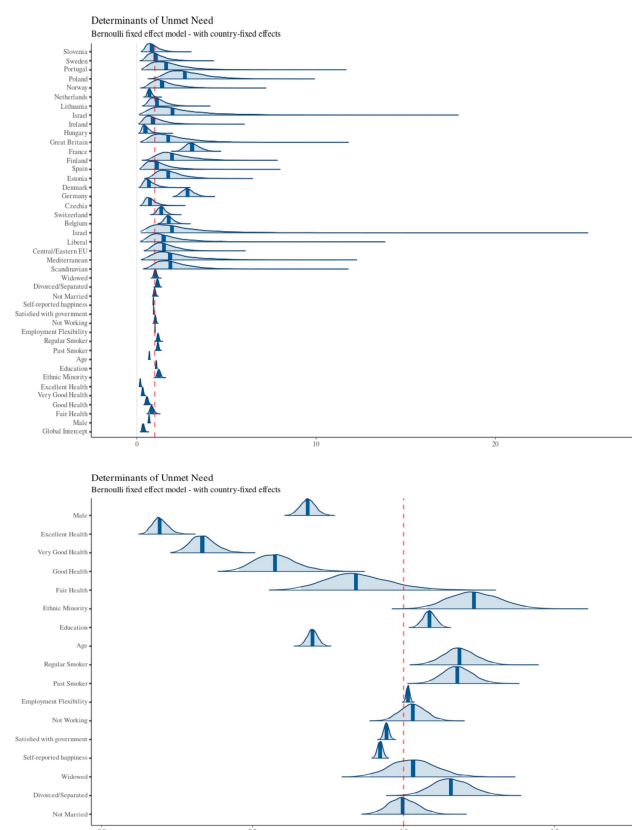


Figure 3: (Above) - Individual-level determinants of unmet medical need, welfare regimes and country-fixed effects. (Below) - A zoom-in on individual-level determinants only from the same model (Bernoulli fixed-effect logistic regression).

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Figure 3 plots the full set of country-fixed-effect posteriors from Model 3 alongside a zoomed-in view of the individual-level estimates from the same model, which remain highly consistent in direction and magnitude with Models 1 and 2.

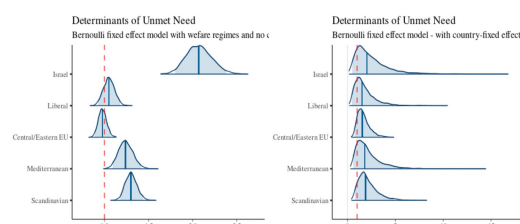


Figure 4: Model 2 (left) and Model 3 (right) with respect to welfare regime posterior distribution results before and after adding country fixed effects

Table 5: Model 4 – Multilevel model of individual determinants of unmet need and welfare regimes on 2nd level (random effects are not exponentiated)

First-level estimates:

	Estimate	Est.Error	Q2.5	Q97.5
Intercept	0.8079	1.263	0.5336	1.456
Male (Female)	0.6923	1.037	0.6444	0.7436
Fair health (Poor)	0.8496	1.131	0.6679	1.081
Good health (Poor)	0.5777	1.121	0.4579	0.7235
Very good health (Poor)	0.3176	1.124	0.2507	0.3988
Excellent health (Poor)	0.174	1.132	0.1359	0.2224
Ethnic minority (No)	1.302	1.066	1.147	1.476
Education years	1.093	1.019	1.054	1.133
Age	0.694	1.025	0.6611	0.729
Past smoker (non-smoker)	1.199	1.042	1.107	1.299
Current smoker (non-smoker)	1.186	1.045	1.089	1.291
Work Flexibility	1.025	1.005	1.014	1.036
Unemployment (no)	1.033	1.041	0.9549	1.117
Government satisfaction	0.929	1.008	0.9148	0.9442
Happiness	0.9339	1.01	0.9165	0.9517
Never married (Married)	1.007	1.047	0.9203	1.1
Divorced/Separated (Married)	1.119	1.054	1.009	1.241
Widowed (Married)	1.037	1.072	0.9019	1.189

Second-level estimates

	Estimate	Est.Error	Q2.5	Q97.5
Conservative	-0.238	0.1977	-0.738	0.0948
Scandinavian	0.023	0.1972	-0.455	0.3541
Mediterranean	-0.022	0.2008	-0.516	0.3168
Central/Eastern EU	-0.258	0.1958	-0.753	0.0704
Liberal	-0.185	0.2007	-0.676	0.1501
Israel	0.471	0.1972	-0.008	0.8204

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Figure 4 compares the welfare-regime posterior distributions before (Model 2) and after (Model 3) adjusting for country-fixed effects, illustrating how the apparent regime effect from Model 2 is substantially reduced once country-level heterogeneity is accounted for directly. Table 5 presents Model 4, the multilevel formulation with welfare regime included as a second-level term with partial pooling; first-level individual estimates remain stable in direction and magnitude relative to the fixed-effect models, while second-level welfare-regime estimates (reported on the untransformed scale) show considerable overlap with zero.

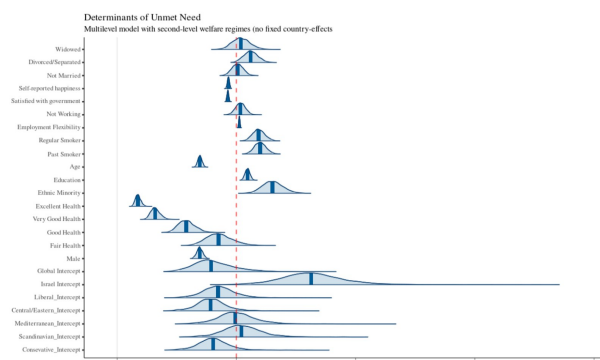


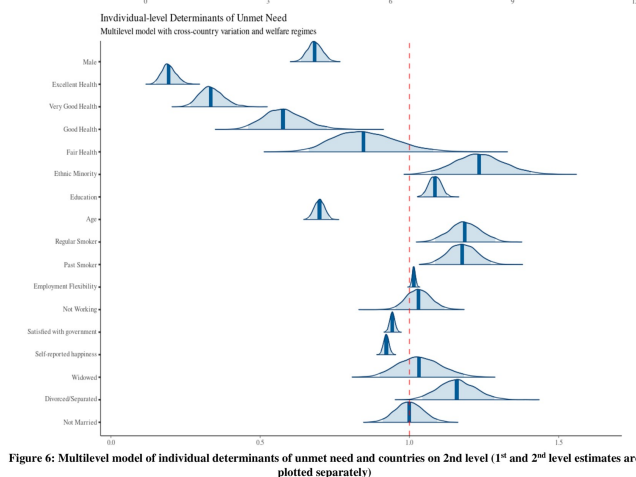
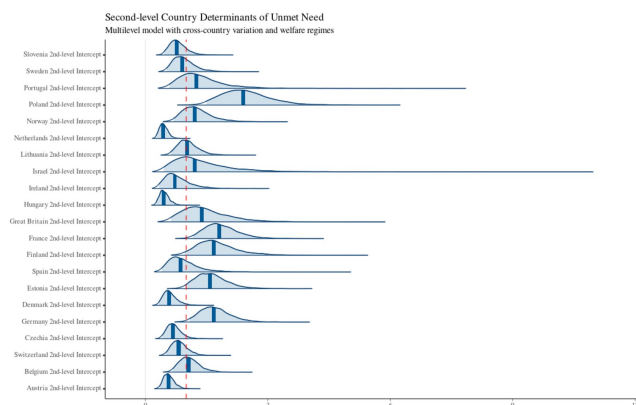
Figure 5: Model 4 - Multilevel model of individual determinants of unmet need and welfare regimes on 2nd level

Table 6: Model 5 – Multilevel model of individual determinants of unmet need and country effects on 2nd level (random intercept estimates are plotted)

First-Level Estimates:				
	Estimate	Est.Error	Q2.5	Q97.5
Intercept	0.614	1.291	0.3755	1.024
Male (Female)	0.6826	1.036	0.6374	0.7309
Fair health (Poor)	0.8459	1.132	0.6624	1.074
Good health (Poor)	0.5759	1.125	0.4603	0.7256
Very good health (Poor)	0.3343	1.127	0.2645	0.4224
Excellent health (Poor)	0.1926	1.134	0.1513	0.246
Ethnic Minority (No)	1.232	1.068	1.083	1.402
Education years	1.085	1.019	1.046	1.126
Age	0.6982	1.025	0.6661	0.7328
Past smoker (non-smoker)	1.177	1.042	1.087	1.276
Current smoker (non-smoker)	1.185	1.045	1.088	1.293
Work Flexibility	1.015	1.006	1.004	1.026
Unemployment (no)	1.031	1.041	0.9526	1.117
Government satisfaction	0.9425	1.009	0.9269	0.9586
Happiness	0.9222	1.01	0.9043	0.9408
Never married (married)	0.9998	1.049	0.9114	1.099
Divorced/Separated (Married)	1.157	1.057	1.039	1.291
Widowed (Married)	1.031	1.073	0.896	1.176
Scandinavian (Conservative)	1.309	1.419	0.6392	2.625
Mediterranean (Conservative)	1.389	1.525	0.5963	3.128
Central/Eastern EU (Conservative)	0.9836	1.374	0.5214	1.861
Liberal (Conservative)	1.101	1.532	0.4745	2.58
Israel (Conservative)	1.893	1.692	0.677	5.3

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Figure 5 visualizes the Model 4 posteriors, including regime-specific second-level intercepts. Table 6 presents Model 5, which adds country effects to the second level alongside welfare regimes; first-level individual-level estimates again remain highly consistent with previous models.

Figure 6: Multilevel model of individual determinants of unmet need and countries on 2nd level (1st and 2nd level estimates are plotted separately)

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Figure 6 shows the second-level country-specific determinants of unmet need from Model 5 (top panel) together with the corresponding individual-level estimates (bottom panel), plotted separately for legibility.

Table 7: Model 6 - Multilevel model of individual determinants of unmet need and welfare regimes on 1st level, country-effects on 2nd level and random slopes for ethnic minority status

First-level Estimates					Second-level Country Intercept				
	Estimate	Est.Error	Q2.5	Q97.5	Estimate	Est.Error	Q2.5	Q97.5	
Intercept	0.6674	1.335	0.3826	1.175	AT	-0.661	0.274	-1.213	-0.136
Male (Female)	0.6861	1.037	0.6381	0.7369	BE	-0.026	0.263	-0.554	0.486
Fair health (Poor)	0.8456	1.135	0.6643	1.094	CH	-0.291	0.275	-0.843	0.249
Good health (Poor)	0.5736	1.127	0.4568	0.7341	CZ	-0.412	0.268	-0.935	0.095
Very good health (Poor)	0.3329	1.129	0.2638	0.4284	DE	0.427	0.255	-0.074	0.931
Excellent health (Poor)	0.1938	1.139	0.1512	0.2521	DK	-0.558	0.311	-1.163	0.056
Education years	1.082	1.019	1.043	1.121	EE	0.458	0.265	-0.084	0.985
Age	0.6961	1.025	0.6624	0.7311	ES	-0.156	0.389	-0.911	0.662
Past smoker (non-smoker)	1.177	1.043	1.085	1.277	FI	0.507	0.309	-0.081	1.132
Current smoker (non-smoker)	1.187	1.046	1.084	1.295	FR	0.498	0.259	0.003	0.990
Work Flexibility	1.015	1.005	1.004	1.025	GB	0.279	0.401	-0.496	1.069
Unemployment (no)	1.024	1.041	0.9471	1.109	HU	-0.855	0.283	-1.430	-0.317
Government satisfaction	0.9415	1.009	0.9256	0.9574	IE	-0.337	0.401	-1.126	0.445
Never married (married)	0.9948	1.048	0.9053	1.09	IL	0.312	0.543	-0.718	1.386
Divorced/Separated (Married)	1.152	1.058	1.035	1.284	LT	-0.100	0.266	-0.642	0.420
Widowed (Married)	1.033	1.073	0.9013	1.183	NL	-0.953	0.279	-1.523	-0.420
Happiness	0.9212	1.01	0.9036	0.939	NO	0.181	0.305	-0.418	0.806
Scandinavian (Conservative)	1.225	1.457	0.581	2.567	PL	0.844	0.263	0.304	1.355
Mediterranean (Conservative)	1.317	1.557	0.548	3.099	PT	0.225	0.390	-0.534	1.018
Central/Eastern EU (Conservative)	0.9386	1.37	0.4975	1.745	SE	-0.096	0.309	-0.700	0.532
Liberal (Conservative)	1.06	1.556	0.4448	2.527	SI	-0.289	0.279	-0.852	0.251
Israel (Conservative)	1.781	1.843	0.534	5.578	Second-level Ethnic Minority Random Slopes				
					Estimate	Est.Error	Q2.5	Q97.5	
					AT	0.359	0.364	-0.377	1.043
					BE	0.300	0.323	-0.339	0.917
					CH	0.318	0.314	-0.318	0.934
					CZ	-0.258	0.459	-1.213	0.599
					DE	0.482	0.218	0.045	0.903
					DK	0.118	0.393	-0.696	0.853
					EE	0.079	0.136	-0.194	0.342
					ES	0.172	0.381	-0.559	0.915
					FI	0.031	0.370	-0.712	0.731
					FR	0.608	0.252	0.118	1.095
					GB	0.321	0.203	-0.073	0.719
					HU	0.538	0.366	-0.194	1.251
					IE	-0.270	0.400	-1.102	0.476
					IL	-0.664	0.178	-1.020	-0.317
					LT	1.027	0.216	0.600	1.440
					NL	0.657	0.316	0.006	1.258
					NO	0.105	0.299	-0.489	0.678
					PL	-0.103	0.449	-0.976	0.799
					PT	-0.077	0.404	-0.889	0.704
					SE	-0.317	0.357	-1.078	0.308
					SI	0.009	0.435	-0.899	0.838

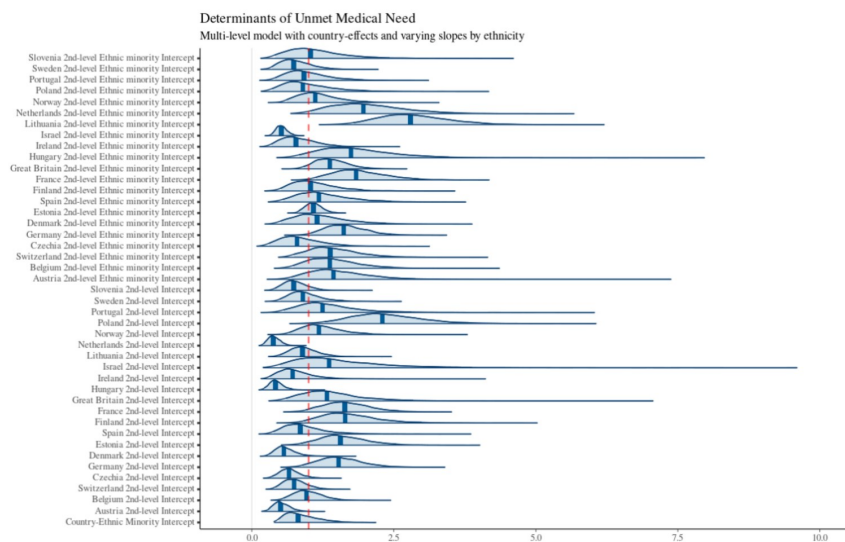


Figure 7: Multilevel model of individual determinants of unmet need and countries on 2nd level (Random effect estimates)

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Table 7 and Figure 7 present our final and most complex specification, Model 6, which allows the second-level country intercepts to vary further by random slopes for ethnic minority status. We found that this affected the slope of a few country-specific intercepts, such that those belonging to ethnic minorities in the Netherlands, Lithuania, Ireland, Hungary, Germany, and France, faced higher excess unmet need compared to ethnic majorities. In other words, the negative effect of ethnic minority status on unmet need for care was more pronounced in these countries compared to the others. Over the last few decades, all of these countries have been accepting of immigration from all ethnicities and creating a multicultural living environment, albeit stratified into the majority and minority ethnic groups. At the same time, they have all faced growth in anti-immigration sentiments and protectionist poli-

cies, which may have left the ethnic minority populations more vulnerable to discrimination and unmet medical need. It may also be that ethnic minority status serves as an imperfect proxy for socioeconomic status and that different European countries, each belonging to a slightly different point on the neoliberal spectrum, vary in their ability and willingness to acknowledge differential need and to support vulnerable subpopulations. Future research should examine cross-country policy differences across a wide range of fields, such as health, immigration, family, or employment to try and pinpoint potential drivers of differential ethnic minority experiences in facing unmet medical need across countries.

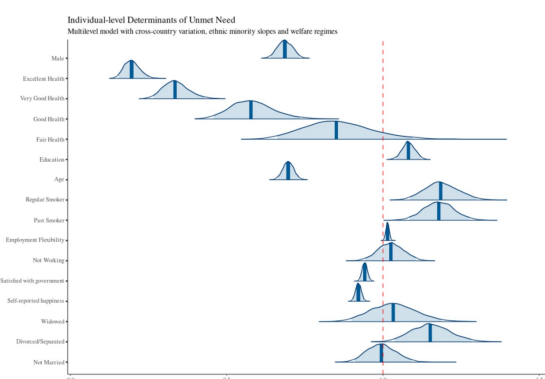


Figure 8: Multilevel model of individual determinants of unmet need and countries on 2nd level (1st level estimates)

From the models above, we can see that the direction and magnitude of the individual-level estimates remains very similar across all models. However, the credible intervals for our estimates become wider as we introduce multiple levels and random slopes. Given these similar effect magnitudes from model to model, we present the results of our best model, Model 6, as determined by the widely applicable information criterion (WAIC), presented in Table 8 below.

Table 8: WAIC criterion for Models 1-6

Type	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Fixed effects, individual-level effects only	24,271.7	23,508.1	24,145.1	24,145.1	23,507.6	23,466.6

Discussion of findings and directions for future research

We found that individual-level sociodemographic characteristics of respondents consistently affect the odds of unmet medical need across all models, whereby the odds are lower for males, ethnic majorities, non-smokers, older individuals, those with good self-reported health, happiness, and those more satisfied with their government, compared to females, ethnic minorities, current and past smokers, younger individuals, those with poor self-reported health and happiness and those who are not satisfied with their government, respectively. Many of these determinants serve as socioeconomic proxies and the results support our initial hypothesis, which expected higher unmet need among those of lower socioeconomic class. Individuals with smoking history could both face higher need due to worse health outcomes, and/or because smoking could be considered a proxy for socioeconomic status. Similarly, ethnic minority status could be one such proxy, but it may also be that those who belong to ethnic minorities face stigma and discrimination, which contributes to their inability to access needed medical services. Females may face higher unmet need than males since they may not have the time to go address their unmet need, having traditionally assumed both formal labor and a higher proportion of informal household chores. The fact that those who self-report being healthier and happier, and who are more satisfied with their government are less likely to face unmet medical need makes intuitive sense.

Surprisingly, we also found that individuals with higher educational attainment and those with higher workplace flexibil-

ity were more likely to face unmet medical need compared to their less educated and less flexible counterparts. This finding was not anticipated by our hypothesis, since it suggests that some aspects of higher socioeconomic status are associated with more unmet need for medical care. While a thorough re-examination of these findings in future research is warranted, we stipulate that this higher unmet need may be driven by higher demand for healthcare services and particularly preventative medicine among the more educated. Since the unmet need is self-reported, it is impossible for us to determine whether it is concentrated among those who have objectively worse health outcomes that require direct unmet medical attention, or among those who have more demand for less pressing needs. Future research should aim to control for objectively reported health status evaluations. It would also be interesting to examine these findings while controlling for the public-private divide of the healthcare system, to see whether the unmet need among the more educated changes depending on whether the respondents' country has an established private healthcare infrastructure, which could satisfy the unmet need among those with higher SES, who may be willing and able to pay for the services.

When it comes to macro-level determinants of unmet need, we found that initially, before controlling for country-fixed effects, welfare regime classification appeared to matter for explaining unmet need, whereby those living in Israel, Scandinavian and Mediterranean regime countries were more likely to face unmet need than those living Conservative countries. However, after adjusting for country-fixed effects, these previously identified effects have disappeared, paving the way for cross-country variation to explain away variation in unmet need instead. This is not particularly surprising, given the numerous limitations associated with our computed welfare regime classification and its inability to fully explain country-level differences with respect to only two policy design aspects: labor de commodification and stratification. Future research should consider constructing more specific macro-level indicators for cross-country policy similarities and differences. These may include evaluative variables of system functionality, such as country-level procedure wait times, number of physicians or hospital beds per capita, or specific policy instruments, such as mandated sick leave or national private/public insurance ratio and how is access to it defined.

In the last multi-level model, we have allowed slopes of our random second-level country intercepts to vary by ethnic minority classification, as discussed above with respect to Table 7 and Figure 7.

4 Challenges and Limitations

While our analysis benefited from a large sample size and statistical power encompassing a large number of countries, the study doesn't come without a fair share of challenges. Some of the variables used in our analysis are limited in their ability to capture the dynamic and ever-changing social context. For example, gender is defined in binary terms, leaving out a potentially significant and informative proportion of individuals who do not self-identify in binary terms without a voice. Sim-

ilarly, due to data availability, ethnicity was only defined in binary terms as belonging to ethnic minority or not, which is limited in its ability to capture the complex effects ethnicity may play in determining unmet need for care. Additionally, questions related to immigration status were not considered in the dataset and were therefore omitted from the regression analyses.

Lastly, due to current constraint on resources and time, we have only used a limited measure of our macro-policy context – welfare regime classification, fully acknowledging that this is an imperfect proxy of varying policy environments that may affect unmet medical need. While our welfare regime variable was constructed after a careful review of social policy literature, we failed to identify a welfare regime category to which Israel would fit. Since a substantial proportion of our sample respondents, 1,971/34,029, or 5.9% of the analytic sample, come from Israel, we have decided to keep it as a separate welfare category, instead of excluding the data or artificially assigning the country to a welfare regime that does not fit the policy environment. Future research should improve upon the methodology of this paper by constructing more detailed covariates for policy differences across countries, such as healthcare affordability (i.e., public/private health insurance variable), or mandated sickness leave generosity.

5 Conclusions

Using Bayesian methods and a set of six logistic regression models, our paper attempted to shed light on the risk-factors associated with facing unmet need for medical care. We identified a set of consistent individual-level sociodemographic risk factors and uncovered clinically significant cross-country variation in unmet medical need. Our results highlight the need to expand healthcare affordability, availability, and accessibility beyond universal interventions and to start targeting high-risk groups within a universal integrated policy system. Specifically, future research should focus on uncovering the mechanisms through which these inequities in unmet medical need arise, and policy makers need to make conscious effort to acknowledge and address the excess unmet medical need among the high-risk groups we identified in this paper.

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Appendix A: Country classification by welfare regime type

Appendix A: Country classification by welfare regime type and the state-market-family division of roles

Welfare Regime	Countries		State Role	Market Role	Family Role
Anglo-Saxon	Great Britain	Ireland	Marginal	Central	Marginal
Continental/ Conservative	Austria Switzerland France	Germany Belgium Luxembourg	Subsidiary	Marginal	Central
Scandinavian	Denmark Sweden Norway	Finland Iceland Netherlands	Central	Marginal	Marginal
Mediterranean	Cyprus Spain Portugal	Italy Turkey Greece	Marginal	Subsidiary	Central
Central/Eastern European	Slovakia Czech Republic Poland Hungary Slovenia	Bulgaria Estonia Croatia Lithuania Romania	Central	Marginal	Subsidiary

Table generated by the author of this paper. Sources: Esping-Andersen 1990, Martin 2015, Beblavy 2008, Haggard and Kaufman 2009